

CAMBIUM ACADEMY • COURSE 01

# Intro to AI

*How AI works, how it is trained, and which model to use*



Free forever. No signup. **cambium ai research institution**

July 2026 edition. The AI world moves fast, so facts carry dates.

WELCOME

# This course was built for beginners



## No background needed

If you can use a web browser, you are qualified.



## No math, no code

We build intuition with pictures and plain English.



## Jargon gets translated

Every technical term arrives with a one-line meaning.

## WHAT YOU NEED



A web browser



About 90 minutes



Zero dollars

## OUTCOMES

# Five things you will walk away with



### **Explain AI in one sentence**

And know how AI, machine learning, and generative AI relate.



### **See how models really work**

Tokens, neural networks, and the transformer, minus the math.



### **Understand training**

From a mountain of text to a helpful assistant, in four steps.



### **Know the 2026 landscape**

Who makes the top models and what makes each one special.



### **Pick the right model**

A practical cheat sheet for writing, coding, research, and more.

## COURSE MAP

# Six short modules, one quiz, one certificate

### 1 What is AI?

10 min

### 2 How models work

12 min

### 3 How models are trained

12 min

### 4 The 2026 model landscape

12 min

### 5 Which model for which job

8 min

### 6 Using AI well

8 min

**Then:** flashcards and the hands-on AI Lab unlock the 20-question quiz. Score 70% or better and print your certificate on the spot.

## MODULE 1

# What is AI?

*Software that learns from examples instead of following hand-written rules.*



# AI in one sentence

*Artificial intelligence is software that **learns patterns from examples** and uses them to make useful guesses about things it has never seen.*



## Traditional software

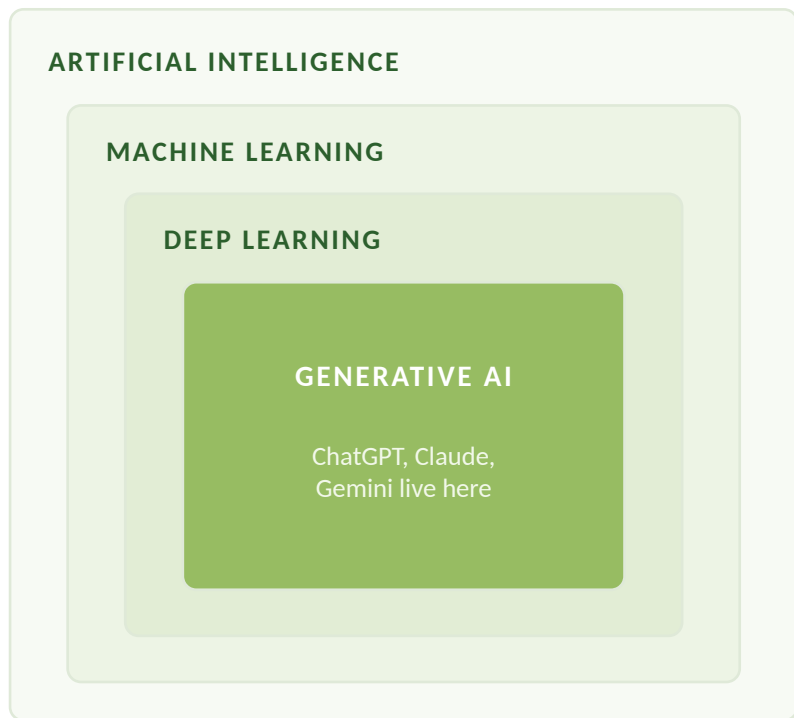
A person writes every rule by hand. "If the email says WIN CASH, mark it as spam." Breaks the moment reality gets messy.



## AI / machine learning

Show it a million spam emails and a million real ones. It finds the telltale patterns itself, including ones no human ever wrote down.

# AI is the family. Generative AI is the newest child.



## Artificial intelligence

Any technique that makes machines act smart, from chess engines to chatbots.

## Machine learning

The dominant technique: learn patterns from data instead of hand-written rules.

## Deep learning

Machine learning with many-layered neural networks. Took off in 2012.

## Generative AI

Deep learning that creates new text, images, audio, and video. The 2022 breakthrough.

# Seventy-five years in one slide

Alan Turing asks: can machines think?

**1950**

Deep learning wins the ImageNet vision contest

**2012**

ChatGPT brings AI to everyone

**2022**

**1997**

Deep Blue beats the world chess champion

**2017**

Google publishes the Transformer paper

**2026**

AI assistants in a billion pockets

*The lesson: AI is not new. What changed recently is scale, data, and a 2017 invention called the transformer.*



# You used AI before breakfast



## Spam filters

ML has guarded your inbox for 20 years



## Phone cameras

Every photo is cleaned up by a neural net



## Maps and traffic

Route predictions learned from millions of trips



## Autocomplete

Your keyboard predicts the next word, LLM-style



## Recommendations

Netflix, YouTube, and Spotify picks



## Voice assistants

Speech to text is deep learning too

*Chatbots are simply the newest face of a technology you already trust daily.*

## MODULE 2

# How AI models work

*A language model is a prediction machine.*



# One trick underneath it all: predict the next word

"The capital of France is \_\_\_\_\_"

**Paris 97%**

Lyon 1% • a 0.5% • ...



## Repeat the trick

Predict a word, add it to the sentence, predict again. A paragraph is thousands of predictions in a row.



## Trained on trillions of words

Good prediction quietly forces the model to absorb grammar, facts, style, and reasoning patterns.



## That is the whole magic

Essays, code, and advice all emerge from one skill: guessing the next word extremely well.

# AI does not read words. It reads tokens.



*one word can be several tokens, and every token is just a number to the model*



## A token is a chunk of text

Roughly three quarters of an English word. 1,000 tokens is about 750 words.



## Models have a token budget

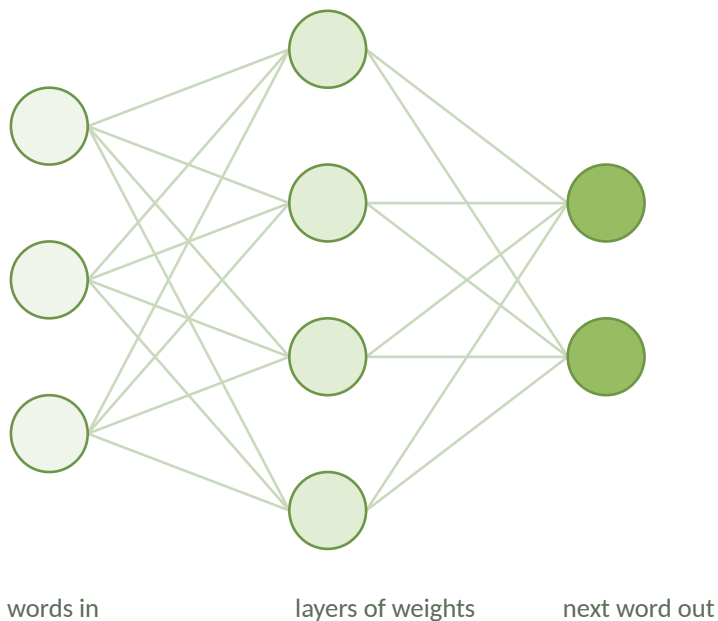
The context window. Fill it up and the oldest content falls out of memory.



## Tokens are the currency of AI

API pricing, speed ratings, and limits are all counted in tokens.

# Billions of tiny dials, tuned by practice



## Every connection is a dial

A weight: one number that gets nudged up or down during training.



## Learning = tuning the dials

Wrong guess? Nudge billions of weights so the right answer scores higher next time.



## Scale is the surprise

Frontier models have hundreds of billions of weights. Somehow, fluency emerges.

# The 2017 invention that changed everything

"The trophy did not fit in the suitcase because **it** was too big." Which "it"? *Attention links "it" to "trophy"*.



## Attention: every word looks at every other word

The model learns which words matter to which, no matter how far apart they sit.



## It reads everything at once

Older AI read word by word. Transformers process a whole text in parallel, so training on internet-scale data became affordable.



## It is the T in GPT

Generative Pre-trained Transformer. Claude, Gemini, and Llama are transformers too.

# The context window: how much it can hold in mind

Typical chat model

200K tokens

*about one long novel*

Claude / Gemini top tiers

1M tokens

*a small bookshelf*

Llama 4 Scout (advertised)

10M tokens

*dozens of books, about half reliably usable in tests*

When the window fills, the earliest content silently falls out. If a long chat gets forgetful, start a fresh one and restate what matters.

## MODULE 3

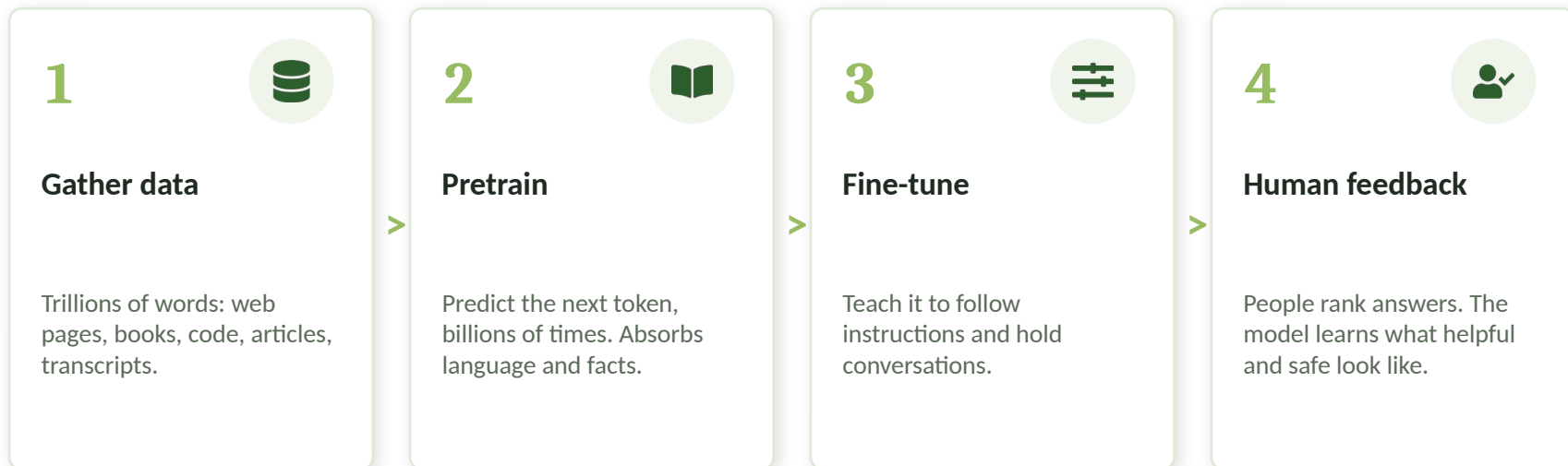
# How models are trained

*Read everything. Learn the job. Learn manners.*





# From raw internet to helpful assistant, in four steps



*Steps 1 and 2 build a brilliant autocomplete. Steps 3 and 4 turn it into an assistant.*

# Why only a handful of labs train frontier models



## Months

of continuous training for one frontier run



## 10,000s

of specialized GPU chips running in parallel



## \$100M+

estimated cost of a single frontier training run



**The efficiency race is on.** In 2026, open models like DeepSeek V4 match top-tier performance at roughly one tenth the serving cost per token, proof that clever engineering can beat raw spending.

# Same brain, before and after finishing school

## BASE MODEL

**A brilliant autocomplete with no manners.**

*You: How do I bake bread?*

*Base model: asked Jim, who had baked for years...*

It just continues the text. It was never taught to answer.

## AFTER FINE-TUNING + FEEDBACK

**A helpful assistant.**



Answers the actual question, step by step



Declines harmful requests



Admits uncertainty instead of bluffing

*This stage is why talking to AI feels natural.*

# Why AI confidently makes things up



## It always answers with the most plausible words

Plausible is not the same as true. There is no built-in fact checker.



## Gaps get filled smoothly

Ask about something it does not know and it may invent names, dates, citations, or numbers that merely sound right.



## Modern tools fight back

Web search, citations, and reasoning modes reduce hallucinations. They do not eliminate them.



**The one rule to remember:** trust, but verify. Check every name, number, date, and citation before you rely on it.

## MODULE 4

# The 2026 model landscape

*A handful of labs, two philosophies.*



# Closed weights or open weights



## Closed weights

You use the model as a service. The recipe stays secret.

**GPT, Claude, Gemini, Grok**

Usually the strongest, easiest to use, and safest defaults.  
You need internet and their servers.



## Open weights

You can download the model and run it yourself.

**Llama, DeepSeek, Qwen, Mistral**

Free, private, customizable, runs offline. You bring the hardware and the elbow grease.

# OpenAI, Anthropic, Google



## OpenAI • GPT

**GPT-5.5 now, GPT-5.6 rolling out July 2026**

The most-used AI on earth. New chips serve replies at up to 750 tokens a second, and an Ultra mode puts teams of AI agents on one hard task.

**Free: ChatGPT free tier**



## Anthropic • Claude

**Sonnet 5 free, Opus 4.8, new Fable 5 on top**

Known for careful long-form work, top-tier coding via Claude Code, and million-token context. Sonnet 5 became the free default on July 1.

**Free: [claude.ai](https://claude.ai)**



## Google • Gemini

**Gemini 3.1 Pro and 3.5 Flash**

Woven through Search, Docs, and Gmail, with the broadest free tier: images, voice, and about five Deep Research reports a month at no cost.

**Free: Gemini app**

# Meta, xAI, DeepSeek, Mistral



## Meta

Muse Spark is its new closed flagship, a strategy shift. Open Llama 4 remains, with a 10M-token context on Scout. Free inside WhatsApp and Instagram.



## xAI • Grok

Grok 4.3 taps the live X firehose for real-time answers, takes video input, and holds a million tokens. Free tier is thin; most features are paid.



## DeepSeek

Open V4 model matches top-tier benchmarks at roughly a tenth of the serving cost. Generous free app. Hosted in China, so mind where your data goes.



## Mistral

Europe's champion. Mistral Large 3 is the largest fully open (Apache 2.0) model from a Western lab. Free casual use via its Vibe chat app.



# The strongest free models now come from open labs

## Qwen 3.5 / 3.6

Alibaba

Open family in every size from 0.8B to 397B parameters. Excellent multilingual skills. Free chat and free weights.

## Kimi K2.6

Moonshot AI

Around a trillion parameters, open, and a favorite for agentic coding on a budget. Free chat at [kimi.com](https://kimi.com).

## GLM-5.2

Z.ai

MIT licensed with a million-token context, rivaling Western flagships on coding benchmarks. Free chat at [chat.z.ai](https://chat.z.ai).

**Worth knowing:** these labs are China-based. The models are world class and free, but chats on their hosted apps fall under Chinese data rules. For sensitive work, run the open weights locally instead.

# Four numbers from 2026

**1B+**

monthly ChatGPT app users by June 2026, the fastest app ever to get there

**900M**

monthly users of Google's Gemini app by May 2026

**5B**

images created with Gemini's Nano Banana tool in about two months

**\$965B**

Anthropic's May 2026 valuation, briefly the highest of any AI lab

# You can run a real AI model on your own computer



## Install Ollama, pick a model

Ollama is a free app for Windows, Mac, and Linux that downloads and runs open models with one command.



## Everything stays local

No internet needed after download. Nothing you type leaves your machine. Cost: zero, forever.



## Small models got good

Qwen3 4B or 8B suits most laptops. Gemma 3 and Phi-4-mini shine on lighter hardware.



**Honest trade-off:** local models are noticeably weaker than the frontier. Perfect for private notes, drafts, and learning. Not for your hardest problems.

## MODULE 5

# Which model for which job?

*There is no single best model. There is a best model for each job.*



# Words and work

| Your job                 | Reach for                             | Also great                                         |
|--------------------------|---------------------------------------|----------------------------------------------------|
| Everyday writing and Q&A | ChatGPT, free tier                    | Claude Sonnet 5, often the more natural writer     |
| Coding                   | Claude with Claude Code               | DeepSeek V4 or Kimi K2.6, strong and free          |
| Research with sources    | Perplexity, free, cites as it answers | Gemini Deep Research, about 5 free reports a month |
| Long documents           | Claude, million-token context         | Gemini 3.1 Pro, 1M context plus Docs and Drive     |

*Snapshot: July 2026. Free tiers and model names shift often; the course page keeps the current version.*

# Media, privacy, and the best free picks

| Your job            | Reach for                           | Also great                                  |
|---------------------|-------------------------------------|---------------------------------------------|
| Images              | Gemini's Nano Banana, free          | Midjourney, paid, for fine art control      |
| Video clips         | Google Veo 3.1, 4K with sound       | Kling 3.0, cheaper per clip                 |
| Voice and speech    | ElevenLabs, free tier               | ChatGPT voice mode, simple and bundled      |
| Private and offline | Qwen3 via Ollama, local             | Gemma 3 or Phi-4-mini on light hardware     |
| Best free overall   | Gemini free tier, broadest features | ChatGPT free tier, easiest to get help with |

# Four questions beat any leaderboard



## Is the data sensitive?

Run an open model locally, or use a paid plan with training turned off.



## Do you need checkable sources?

Use a research tool that cites: Perplexity or Deep Research.



## Is the input huge?

Pick a million-token context model: Claude or Gemini.



## Just everyday help?

Any of the big three free tiers is excellent. Stop optimizing.

**Best advice:** pick one model and learn it deeply. Prompting skill transfers everywhere; tool-hopping does not compound.

## MODULE 6

# Using AI well

*The human stays in charge.*





# Five golden rules



## **Verify before you rely**

Check names, numbers, dates, and citations. AI is confident even when wrong.



## **Guard private data**

Do not paste passwords, client records, or secrets into tools you have not vetted.



## **You own the judgment**

AI drafts, suggests, and accelerates. Decisions and responsibility stay with you.



## **Disclose when it matters**

School, work, publishing: say when AI helped. Honesty ages well.



## **Reassess monthly**

Models leapfrog each other constantly. Today's winner is next month's runner-up.

# Better answers in four moves

## Role

Who should the AI be?

## Context

What does it need to know?

## Task

Exactly what do you want?

## Format

How should the answer look?

### EXAMPLE PROMPT

**You are a seasoned travel planner.** Two adults are visiting Boise for three days in October on a modest budget. **Plan a day-by-day itinerary with one outdoor highlight per day.** Present it as a simple table.

*Role and task in green, context and format in black. Course 02 goes deep on this next week.*

# Hand-verified free resources



## How Machines Learn, CGP Grey

The best 5-minute mental model on YouTube



## Intro to Large Language Models, Andrej Karpathy

A legendary 1-hour talk by a top AI researcher



## Neural Networks, 3Blue1Brown

Beautiful visual series on what is inside the box



## Elements of AI, University of Helsinki

The classic free intro course, 2M+ learners



## AI for Everyone, Andrew Ng

Free-to-audit course made for non-engineers

Every link, with time estimates and levels, lives on the course resources page. Each one was verified live in July 2026.

YOUR TURN



# Finish the path. Earn your certificate.



Step 1, the lecture: done, you just finished it



Step 2: clear all 24 flashcards until they stick



Step 3: explore the three stations in the AI Lab



The quiz unlocks: 20 questions, 70% to pass, certificate prints on the spot

**Next week • Course 02: Prompting Essentials**

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